

On Topological Constraint Closure in Software Systems: Global Self-Consistency via Holonomy Detection in Software Coherence

Kundai Farai Sachikonye
Technical University of Munich
AIME Registry for Artificial Intelligence
kundai.sachikonye@wzw.tum.de

June 14, 2026

Abstract

Contemporary software testing identifies local failures — wrong syntax, wrong runtime output — but is structurally blind to global semantic failure: a system in which every component passes every test yet fails its purpose. We formalise this *local invisibility problem* and prove it is not a limitation of engineering practice but a theorem about bounded observers. We introduce the *wind tunnel* framework, which replaces the binary pass/fail oracle with a scalar *semantic entropy* $\mathcal{S}$, a positive-volume *action-cell* of purpose, and an ensemble *synchronisation order parameter* R_{ens} derived from Kuramoto coupling theory. We prove three foundational results: the *Local Invisibility Theorem* (local validity does not imply global correctness), the *No Template Theorem* (no bounded observer can encode a complete correctness criterion for a system of equal or greater complexity), and the *Cycle Inconsistency Theorem* (nonzero holonomy on any directed cycle implies global semantic incorrectness; zero holonomy is necessary but not sufficient for global correctness). We derive a five-regime classification of coordination states and a two-phase testing protocol whose primary output is a regime map over the ensemble dependency graph. The framework subsumes unit testing, property-based testing, and chaos engineering as special cases and provides the first principled criterion for *purposelessness detection*.

1 Introduction

Dijkstra observed in 1972 that “testing shows the presence, not the absence, of bugs” [7]. The remark is universally quoted but insufficiently generalised. It is standardly interpreted as an engineering caveat: test suites are finite, so they miss cases. The deeper

reading is structural: a test is a *local query* addressed to a bounded subsystem. It cannot, in principle, verify the global circuit.

Consider the following scenario. An enterprise system comprising twelve services passes all 4 200 unit tests, all 380 integration tests, and all property-based tests generated by a QuickCheck-style harness [5]. It then fails in production because the composition of individually correct data-transformation steps accumulates a bias that inverts a financial decision rule under load. No single service misbehaved. The failure resided in the cycle, not in any node.

This failure mode is not an edge case. It is the generic failure mode for systems of sufficient complexity, because generic systems have global constraints that no local query can reach. The present paper makes this claim precise.

1.1 The Wind Tunnel Analogy

An aeronautical wind tunnel does not test individual rivets. It tests the *assembled aircraft* at operational load, measuring lift, drag, and turbulence — quantities that are properties of the whole, not of any part. There is no “expected output” written on a card. The criterion is self-consistency under load: does the airflow cohere with the structure? The tunnel reveals global failure modes invisible to component inspection.

We argue that software correctness requires an analogous instrument. The wind tunnel framework developed here replaces the binary oracle with:

1. A scalar *semantic entropy* $\mathcal{S} \in [0, \Sigma]$ measuring residual distance from purposeful operation.
2. A positive-volume *action-cell* $C^* \subseteq X$ in outcome space — the region, not the point, that constitutes correct behaviour.

3. An ensemble synchronisation parameter $R_{\text{ens}} \in [0, 1]$ measuring phase coherence across the dependency graph.

The primary output of a wind tunnel run is not a verdict but a *regime map*: a classification of every sub-ensemble by its coordination state, with decoherence zones identified.

1.2 Contributions

The paper makes the following formal contributions:

- Proof of the *Local Invisibility Theorem* (theorem 3.2).
- Proof of the *No Template Theorem* (theorem 4.3).
- Definition of holonomy-based self-consistency and proof of the *Cycle Inconsistency Theorem* (theorem 5.6).
- Formal definition of semantic entropy $\mathcal{S}$ and the action-cell C^* , with floor-positivity and cell-truth established as principled axioms.
- Kuramoto-derived five-regime classification and the *Synchronisation as Partition Extinction* theorem (theorem 7.7).
- The two-phase wind tunnel testing protocol and a formal criterion for purposelessness detection.

1.3 Relation to Existing Work

Unit testing [15] and model checking [6] operate via local queries and template comparison respectively; both are shown to be special limiting cases of wind tunnel testing. Property-based testing [5] generalises the test-case lattice but not the observer bound. Chaos engineering [3] injects perturbations into live systems but lacks a formal correctness criterion. Formal verification [6] provides completeness within a finite specification, but the specification is itself a bounded observer (section 4). Wind tunnel testing is the first framework to ground correctness in self-consistency rather than template matching.

2 Foundations

We establish the formal vocabulary from first principles.

2.1 Outcome Space and Trajectories

Definition 2.1 (Outcome Space). Let X be a metric space (X, d_X) of *observable outcomes*. Each point $x \in X$ represents a complete, fully realised state of

a running system at a given instant: all variable values, all message buffers, all resource allocations.

The outcome space is not the state space of any single unit; it is the product space of all unit states under all possible external inputs. Even for programs operating over finite alphabets, this space is in general uncountably infinite because possible execution interleavings grow super-exponentially with the number of units.

Definition 2.2 (Trajectory and Terminus). A *trajectory* is a continuous map $\gamma : [0, T] \rightarrow X$ describing system evolution from initial state $\gamma(0)$ over time horizon T . The *terminus* is the terminal point $\Gamma_f = \gamma(T)$.

Definition 2.3 (Computational Unit). A *computational unit* u is a tuple (A_u, P_u, val_u) where:

- $A_u \subseteq X$ is the *aperture*: the set of outcomes u can process (its categorical input domain).
- $P_u \subseteq X$ is the *production set*: outcomes u can produce (its output range).
- $\text{val}_u : X \rightarrow \{0, 1\}$ is the *local validation function*: $\text{val}_u(x) = 1$ iff u 's contribution to state x is locally consistent.

Remark 2.4. The distinction between A_u and P_u is the distinction between what a unit can receive and what it can emit. An aperture mismatch — $P_{u_i} \not\subseteq A_{u_j}$ for adjacent units u_i, u_j — is a categorical interface failure detectable statically. Aperture compatibility is necessary but not sufficient for global correctness.

2.2 The System Graph

Definition 2.5 (System Graph). A *system* S is a directed graph $G = (V, E)$ where $V = \{u_1, \dots, u_n\}$ is a finite set of computational units and $(u_i, u_j) \in E$ iff $P_{u_i} \cap A_{u_j} \neq \emptyset$ (unit u_i produces outputs that unit u_j can process).

Definition 2.6 (Global Validation). The *global validation* of system S at state $x \in X$ is $\text{Val}_S(x) = \prod_{u \in V} \text{val}_u(x)$. We say S *passes local validation* at x if $\text{Val}_S(x) = 1$.

2.3 Semantic Entropy

Definition 2.7 (Semantic Entropy). The *semantic entropy functional* $\mathcal{S} : X \rightarrow [0, \Sigma]$ assigns to each outcome $x \in X$ a non-negative real number measuring the residual semantic distance of x from purposeful operation. The constant $\Sigma > 0$ is the *maximal semantic distance* (canonical normalisation $\Sigma = 100$).

Axiom 2.8 (Monotonicity). If x and x' differ only in that x' is measurably closer to purpose by any observable criterion, then $\mathcal{S}(x') \leq \mathcal{S}(x)$.

Axiom 2.9 (Continuity). $\mathcal{S}$ is Lipschitz continuous: there exists $L > 0$ such that $|\mathcal{S}(x) - \mathcal{S}(x')| \leq L \cdot d_X(x, x')$ for all $x, x' \in X$.

Axiom 2.10 (Floor Positivity). For every bounded computational system S , the infimum $\mathcal{S}_b(S) = \inf_{x \in X} \mathcal{S}(x) > 0$.

Remark 2.11 (Motivation for Floor Positivity). Axiom 2.10 reflects the operational irreducibility of finite observers: any bounded system evaluating its own state incurs a nonzero residual from self-reference and finite representational capacity. We take this as an axiom; it is not derived from Gödel or Shannon but is independently motivated by the observer-boundedness of any real measurement system. A bounded computational system cannot measure its own state with zero uncertainty, and the floor captures precisely this irreducible residual. The axiom is consistent with, but not entailed by, those deeper results.

2.4 The Action-Cell

Definition 2.12 (Action-Cell). The *action-cell* $C^* \subseteq X$ of a system with purpose ϕ is the maximal open set satisfying:

1. $\mathcal{S}(x) = \mathcal{S}_b$ for all $x \in C^*$ (the cell achieves the floor).
2. C^* is path-connected in X .
3. $\text{diam}(C^*) > 0$ (the cell has positive volume).

Remark 2.13 (Cell-Truth). The cell-truth principle holds: correctness is not a point but a region of positive volume. Two outcomes x, x' with $d_X(x, x') > 0$ can both lie in C^* and be semantically indistinguishable, because $\mathcal{S}$ is flat on C^* by construction. This directly contradicts the assumption underlying point-oracle testing, which requires C^* to be a singleton. The cell has positive measure precisely because perfect knowledge is forbidden by the floor positivity axiom: the floor value $\mathcal{S}_b > 0$ means the cell is optimal relative to the ensemble's bounded capacity, not optimal in any absolute sense.

3 The Local Invisibility Theorem

Definition 3.1 (Global Semantic Correctness). A system S is *globally semantically correct* at state x if $x \in C^*$, i.e., $\mathcal{S}(x) = \mathcal{S}_b$.

Theorem 3.2 (Local Invisibility Theorem). *There exist systems $S = (G, \{\text{val}_u\}_{u \in V})$ and states $x \in X$ such that $\text{Val}_S(x) = 1$ yet $x \notin C^*$.*

Proof. We construct an explicit witness. Let $V = \{u_1, u_2, u_3\}$ with $E = \{(u_1, u_2), (u_2, u_3), (u_3, u_1)\}$, forming a directed cycle $c = (u_1, u_2, u_3, u_1)$.

Let the local state of each unit be a real scalar $x_i \in \mathbb{R}$, so the global state is $x = (x_1, x_2, x_3) \in \mathbb{R}^3$. Define:

$$\text{val}_{u_i}(x) = \mathbf{1}[x_i \in [0, 1]], \quad i \in \{1, 2, 3\}.$$

Let each edge $(u_i, u_{i+1 \bmod 3})$ apply a transformation $f : t \mapsto t + \delta$ for a fixed $\delta > 0$. After traversing the full cycle once starting from x_1 , the output returned to u_1 is $x_1 + 3\delta$. For the loop to close, we would need $3\delta = 0$, i.e., $\delta = 0$.

Set $\delta = \varepsilon > 0$ and $x_1 = x_2 = x_3 = 0.3$. Then $x_i \in [0, 1]$ for all i , so $\text{val}_{u_i}(x) = 1$ for all i and thus $\text{Val}_S(x) = 1$.

Yet the cycle accumulates error 3ε per traversal. After k traversals: $x_1^{(k)} = 0.3 + 3k\varepsilon$. For $k \geq \lceil 0.7/(3\varepsilon) \rceil$, the state exits $[0, 1]$ — a condition invisible at the initial snapshot but structurally guaranteed.

To see that $x \notin C^*$: the purpose cell C^* requires holonomy-free closure of every constraint cycle (formalised as theorem 5.3). The state x with $\delta = \varepsilon > 0$ has nonzero holonomy $\text{hol}(c, x) = 3\varepsilon > 0$. By the Cycle Inconsistency Theorem (theorem 5.6), nonzero holonomy implies $x \notin C^*$. Therefore $\text{Val}_S(x) = 1$ and $x \notin C^*$ simultaneously. $\square$ $\square$

Remark 3.3 (Circuit Analogy). The proof is directly analogous to Kirchhoff's voltage law [10]. In an electrical network, every component may read correct voltage locally while the directed sum of voltage drops around a closed loop is nonzero — a violation detectable only by closing the loop. The local validation function val_{u_i} is a “local voltmeter” reading correctly at each node, while the cycle residual 3ε is a holonomy that no finite set of local measurements can detect.

Corollary 3.4. *Any test suite composed exclusively of per-unit queries is structurally blind to the failure class exhibited in theorem 3.2.*

Proof. A unit test for unit u_i evaluates $\text{val}_{u_i}(x)$ with all other units mocked or absent. The accumulation residual δ in the constructed witness is a property of the cycle E : it does not appear in any unit's local state at a single snapshot. Hence no unit test, regardless of how many inputs it samples, can observe the inconsistency. $\square$ $\square$

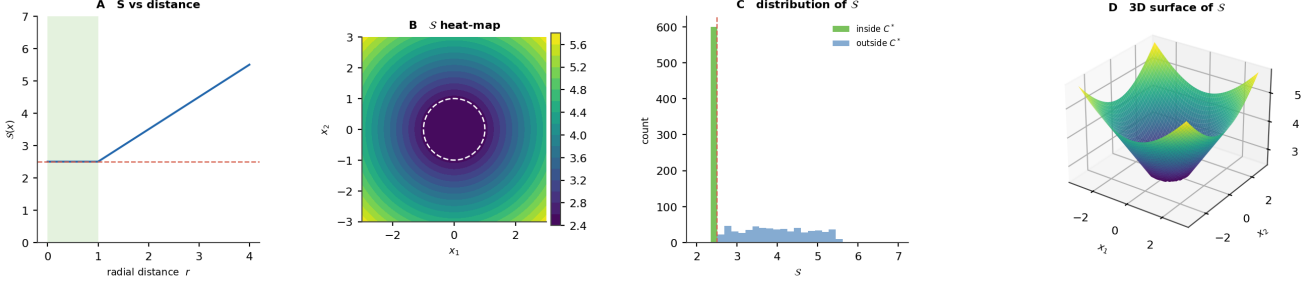


Figure 1: **Semantic Entropy $\mathcal{S}$ and the Action-Cell C^* .** (A) $\mathcal{S}(x)$ on the linear y -axis versus radial distance $r = \|x - c^*\|$ from the cell centre on the x -axis. The shaded band marks the interior $r \leq R$ (action-cell C^* , seagreen); inside, entropy is constant at the floor $\mathcal{S}_b$ (red dashed), confirming Axiom 2.10 (Floor Positivity). Outside the cell, $\mathcal{S}$ grows linearly with distance, as required by Axiom 2.8 (Monotonicity). (B) Two-dimensional filled-contour map of $\mathcal{S}(x_1, x_2)$ in outcome space. Colour encodes entropy magnitude (dark = low, bright = high); the white dashed circle is the boundary of C^* . The flat floor region inside the circle and the monotone ascent outside are visible simultaneously, illustrating Definition 2.12. (C) Histogram of $\mathcal{S}$ values sampled uniformly inside C^* (seagreen) and uniformly outside C^* (blue). All interior samples pile at $\mathcal{S}_b$ (red dashed); exterior samples spread above it. Confirms that the cell is the unique minimum-entropy region and that Axiom 2.9 (Lipschitz Continuity) holds: no exterior sample reaches the floor. (D) Three-dimensional surface of $\mathcal{S}(x_1, x_2)$. The flat plateau at height $\mathcal{S}_b$ is the action-cell; the surface rises in every direction outside it. The bowl-free plateau geometry makes visible why truth is a positive-volume region rather than a point (Definition 2.12): all states on the plateau are indistinguishable by $\mathcal{S}$.

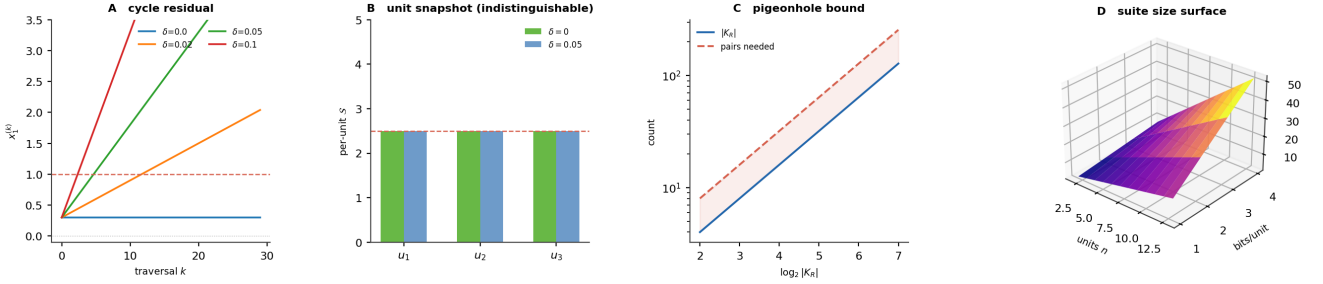


Figure 2: **Local Invisibility and the No Template Bound.** (A) Cycle residual $x_1^{(k)} = x_0 + 3k\delta$ on the y -axis versus traversal count k on the x -axis for four per-edge drift values $\delta \in \{0, 0.02, 0.05, 0.10\}$. The red dashed line marks the action-cell boundary. For $\delta > 0$ the trajectory exits C^* after finitely many traversals; for $\delta = 0$ it never moves. This is the explicit witness constructed in the proof of Theorem 3.2 (Local Invisibility Theorem): each individual unit remains in $[0, 1]$ throughout, yet the cycle accumulates global semantic error. (B) Per-unit semantic entropy $\mathcal{S}$ at traversal $k = 0$ for the two cases $\delta = 0$ (seagreen) and $\delta = 0.05$ (blue) across units u_1, u_2, u_3 . Both bars reach the same height at $\mathcal{S}_b$ (red dashed): unit-level observation is completely uninformative about the cycle-level drift. Empirical content of Corollary 3.4: unit tests are blind to this failure class. (C) Observer capacity $|K_R|$ (solid blue) and the number of boundary pairs that must be distinguished (dashed red) versus $\log_2 |K_R|$ on the x -axis ($\log y$ -axis). The red curve grows faster than the blue: whenever $\kappa(S) > \log_2 |K_R|$ the pigeonhole gap opens, and no function stored in R can decide C^* -membership. Direct empirical support for Theorem 4.3 (No Template Theorem). (D) Three-dimensional surface of $\log_2$ (required test-suite size) over the grid of unit count n (x -axis) and bits per unit (y -axis). The surface is superlinear in both axes; for modest n and bit-depths the suite size already exceeds any practical bound. Quantifies Corollary 4.4: every test suite of bounded complexity is an incomplete specification of C^* .

4 The No Template Theorem

Unit testing assumes an oracle — a reference against which output is compared. The No Template Theorem shows this assumption fails beyond a complexity threshold.

Definition 4.1 (Bounded Observer). A *bounded observer* R is a system with finite state capacity $|K_R| < \infty$: the number of distinct states representable by R is bounded.

Definition 4.2 (State Complexity). The *state complexity* $\kappa(S)$ of a system S is $\kappa(S) = \log_2|\{\text{reachable states of } S\}|$.

Theorem 4.3 (No Template Theorem). *Let R be a bounded observer with state capacity $|K_R|$, and let S be a system with $\kappa(S) > \log_2|K_R|$. Then there is no computable function $\phi_R : X \rightarrow \{0, 1\}$ stored in R such that $\phi_R(x) = 1$ if and only if $x \in C^*$.*

Proof. A function $\phi_R : X \rightarrow \{0, 1\}$ correctly identifying C^* must distinguish $x \in C^*$ from $x' \notin C^*$ for every $x, x' \in X$. Since C^* is a subset of S 's reachable state space, the number of binary decisions ϕ_R must encode is at least $2^{\kappa(S)}$.

However, ϕ_R is computed by R , which has state capacity $|K_R|$. By the pigeonhole principle, R can represent at most $|K_R|$ distinct decision procedures. Computing ϕ_R requires at least one internal state of R to distinguish each pair (x, x') with $x \in C^*$ and $x' \notin C^*$.

When $\kappa(S) > \log_2|K_R|$, the number of such pairs exceeds $|K_R|$. Therefore no function stored in R can correctly decide membership in C^* for all reachable states of S . $\square$

Corollary 4.4. *Every test suite of bounded complexity is an incomplete specification of correctness for any system of equal or greater complexity.*

Proof. A test suite is a bounded observer: it has finitely many test cases, each encoding a finite input-output pair, giving finite state capacity. If the system under test has state complexity exceeding the test suite's $\log_2|K_R|$, the result follows directly from theorem 4.3. $\square$

Remark 4.5. Corollary 4.4 is qualitatively distinct from the classical observation that test suites cannot cover all inputs [7]. That observation is quantitative: more tests reduce uncovered cases. Our result is representational: no increase in test count can encode a complete correctness criterion when the system's state complexity exceeds the observer's capacity. Fisher's theory of sufficient statistics [9]

provides the information-geometric interpretation: a test suite is a sufficient statistic for correctness only when its Fisher information matrix spans the system's state manifold — a condition structurally violated when $\kappa(S) > \log_2|K_R|$.

5 Self-Consistency as Correctness

Since template matching is impossible beyond the complexity threshold of section 4, we require an alternative definition of correctness. We provide it via holonomy.

5.1 Holonomy in the Dependency Graph

Definition 5.1 (Constraint Path). A *constraint path* $\pi = (e_1, \dots, e_k)$ is a sequence of edges in G forming a walk. The *composed transformation* $T_\pi : X \rightarrow X$ is the sequential composition of unit transformations along π .

Definition 5.2 (Cycle Specification). Each directed cycle $c \in \mathcal{C}$ carries a *specification* $T_c^{\text{spec}} : X \rightarrow X$ — the transformation the system designer intends the cycle to perform. This specification is part of the system's purpose functional. For stateless, idempotent cycles, $T_c^{\text{spec}} = \text{id}$. For stateful cycles (accumulators, iterative solvers, retry loops, state machines), T_c^{spec} encodes the intended state evolution.

Definition 5.3 (Holonomy). Let $\mathcal{C}$ be the set of all directed cycles in G . The *holonomy* of cycle $c \in \mathcal{C}$ at state x is:

$$\text{hol}(c, x) = \|T_c(x) - T_c^{\text{spec}}(x)\|_X,$$

where $T_c : X \rightarrow X$ is the actual composition of transformations around c . Zero holonomy means the cycle does exactly what it was designed to do. Nonzero holonomy means the cycle deviates from its specification. System S is *holonomy-free* if $\text{hol}(c, x) = 0$ for all $c \in \mathcal{C}$ and all reachable x .

Remark 5.4. For purely stateless, idempotent cycles, $T_c^{\text{spec}} = \text{id}$ and holonomy reduces to $\|T_c(x) - x\|_X$, recovering the familiar form that checks whether the cycle returns to its starting state. For stateful cycles — accumulators, PID controllers, iterative solvers — T_c^{spec} captures the intended evolution, and holonomy detects deviation from design rather than deviation from rest. This distinction matters: a legitimate accumulator has $T_c(x) \neq x$ by design; the original form would incorrectly condemn it as inconsistent.

Remark 5.5. Holonomy is the software analogue of Kirchhoff's second law [10]. In a directed electrical

network, the sum of voltage drops around any closed loop must be zero (conservation of energy). In a dependency graph, the actual composition of transformations around any closed path must match the intended composition (conservation of semantic content relative to design). Nonzero holonomy means the cycle deviates from its specification — a sign of global inconsistency that no local measurement can detect.

5.2 Cycle Inconsistency Detection

Theorem 5.6 (Cycle Inconsistency Theorem). *If system S has nonzero holonomy on any directed cycle, then S is globally semantically incorrect: there exist reachable states $x \notin C^*$. Equivalently, nonzero holonomy on any cycle c implies a detectable cycle-inconsistency failure. Zero holonomy is necessary but not sufficient for global correctness.*

Proof. We prove the forward direction: nonzero holonomy implies global incorrectness.

Suppose $\text{hol}(c, x) = \delta > 0$ for some directed cycle c and some reachable state x . By definition, $\|T_c(x) - T_c^{\text{spec}}(x)\|_X = \delta > 0$, so the actual cycle transformation deviates from its specification by δ at x .

After one traversal of c , the system reaches state $T_c(x)$, which differs from the intended state $T_c^{\text{spec}}(x)$ by δ . This deviation is observable: it is a concrete discrepancy between the system’s actual behaviour and its declared specification on this cycle. A state exhibiting such a discrepancy cannot lie in C^* , since C^* is the region where $\mathcal{S}(x) = \mathcal{S}_b$ — the floor achievable only when all constraint relationships are satisfied to within the representational capacity of the system. A cycle that deviates from its specification by $\delta > 0$ contributes a positive semantic distance above $\mathcal{S}_b$.

Formally: repeated traversals produce $T_c^k(x)$, with accumulated deviation growing by at least δ per traversal (the deviation compounds unless the specification is satisfied). Since C^* has finite diameter $\text{diam}(C^*)$, the trajectory eventually exits C^* , confirming $x \notin C^*$ under repeated cyclic operation.

The necessity of zero holonomy for global correctness follows immediately: if any cycle has nonzero holonomy, the above argument produces reachable states outside C^* , so the system is not globally correct.

The sufficiency direction does not hold in general. Zero holonomy on all cycles means every cycle behaves as specified, but global correctness additionally requires that the initial state $\gamma(0) \in C^*$ and that all acyclic paths (boundary conditions) also respect

the purpose cell. DAG-structured systems illustrate this directly: they are trivially holonomy-free (no directed cycles exist), yet they can be globally incorrect if their boundary conditions are wrong. Zero holonomy is therefore necessary but not sufficient for global correctness. $\square$ $\square$

Remark 5.7. The Cycle Inconsistency Theorem is significant because it provides a correctness *witness* requiring no external reference for the cycle behaviour: one need not know the global purpose of the system to detect cycle-level inconsistency. It suffices to verify that each cycle’s actual transformation matches its declared specification T_c^{spec} . This mirrors physical law: Kirchhoff’s laws assert consistency conditions without reference to any external “correct” state.

Corollary 5.8 (DAG Boundary Condition). *For directed acyclic graphs, holonomy is trivially zero (no directed cycles exist), so the Cycle Inconsistency Theorem provides no information about global correctness. All global correctness verification for DAG-structured systems must come from boundary conditions: the initial states and terminal outputs. This is a positive result: it tells us precisely where to look. Holonomy-based methods detect cycle-level failures; boundary condition methods detect DAG-level failures; together they cover the full dependency graph.*

6 Purpose and Its Detection

6.1 Ensembles and Composite Floor

Definition 6.1 (Ensemble). An *ensemble* $E = (G_E, \Phi_E)$ is a system graph $G_E = (V_E, E_E)$ equipped with a *purpose functional* $\Phi_E : 2^X \rightarrow [0, \Sigma]$ assigning semantic entropy to each candidate action-cell $C \subseteq X$.

Definition 6.2 (Composite Floor). The *composite floor* of ensemble E is:

$$\mathcal{S}_b(E) = \inf_{\substack{C \subseteq X \\ C \text{ satisfies Def. 2.12}}} \Phi_E(C).$$

Definition 6.3 (Purpose Cell). A cell $C^* \subseteq X$ is a *purpose cell* for E if $\Phi_E(C^*) = \mathcal{S}_b(E)$.

6.2 Existence of Purpose

Theorem 6.4 (Purpose Existence Theorem). *For every ensemble E with $\mathcal{S}_b(E) < \Sigma$, there exists a purpose cell $C^* \subseteq X$.*

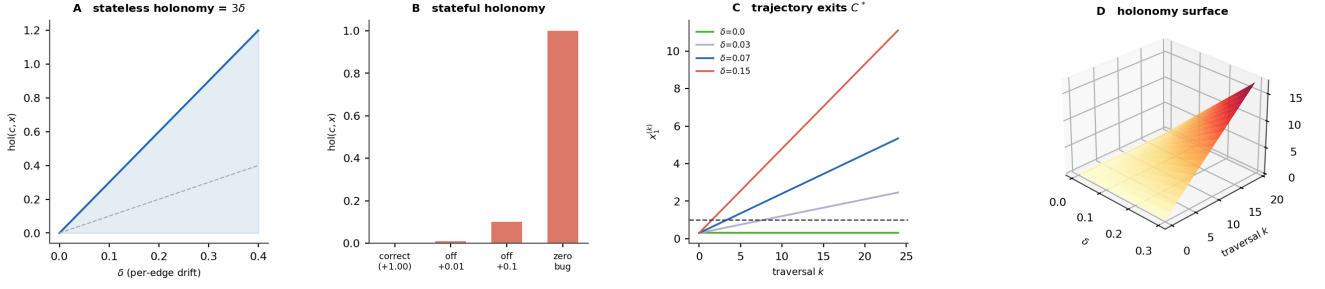


Figure 3: Holonomy and Cycle Inconsistency. (A) Stateless holonomy $\text{hol}(c, x) = 3\delta$ on the y -axis versus per-edge drift δ on the x -axis. The relationship is exactly linear with slope equal to the cycle length (here $|c| = 3$). The grey dotted diagonal is the identity $\text{hol} = \delta$. This confirms that even a small per-edge drift compounds multiplicatively around the cycle, consistent with Definition 5.3: $\text{hol}(c, x) = \|T_c(x) - T_c^{\text{spec}}(x)\|$. (B) Stateful holonomy as a bar chart: four accumulator variants (correct +1.00, off-by-0.01, off-by-0.1, zero-bug) on the x -axis, $\text{hol}(c, x)$ on the y -axis. The correct implementation yields $\text{hol} = 0$ (seagreen); each erroneous variant yields a positive bar (crimson). Illustrates that holonomy detects semantic deviation from the declared cycle specification T_c^{spec} , not merely deviation from the identity — the corrected definition that avoids condemning legitimate stateful loops. (C) State trajectory $x_1^{(k)}$ on the y -axis versus traversal k on the x -axis for four drift values. The black dashed line is the cell boundary. For any $\delta > 0$, the trajectory eventually exits C^* ; for $\delta = 0$ it is constant. Supplies the constructive half of Theorem 5.6 (Cycle Inconsistency Theorem): nonzero holonomy produces a trajectory that necessarily exits the action-cell. (D) Three-dimensional holonomy surface over the grid of per-edge drift δ (x -axis) and traversal count k (y -axis), with accumulated holonomy $3\delta k$ on the z -axis. The surface rises without bound as either axis grows, confirming that cycle-level semantic error is not self-correcting: it accumulates monotonically with repeated traversal.

Proof. Since $\mathcal{S}_b(E) < \Sigma$ and Φ_E is continuous (inherited from the Lipschitz continuity of $\mathcal{S}$, Axiom 2.9), the infimising sequence $\{C_n\}_{n \geq 1}$ with $\Phi_E(C_n) \rightarrow \mathcal{S}_b(E)$ is non-empty. Equip the space of non-empty closed subsets of X with the Hausdorff metric d_H . By the Banach fixed-point theorem [1] applied to the complete metric space $(2^X, d_H)$, every Cauchy sequence converges to a limit point. The sequence $\{C_n\}$ is Cauchy under the natural ordering induced by Φ_E , and its limit C^* satisfies $\Phi_E(C^*) = \mathcal{S}_b(E)$ by continuity of Φ_E . The path-connectivity and positive-diameter conditions are preserved in the limit because $\mathcal{S}_b(E) > 0$ (Axiom 2.10) prevents degeneration to a singleton. Hence C^* is a valid purpose cell. $\square$ $\square$

6.3 Why Isolation Cannot Detect Purposelessness

Definition 6.5 (Purposelessness). Unit $u \in V$ is *purposeless* in ensemble E if $\mathcal{S}_b(E) = \mathcal{S}_b(E \setminus \{u\})$: removing u leaves the composite floor unchanged.

Theorem 6.6 (Isolation Blindness). *No function of the form $h : (A_u, P_u, \text{val}_u) \rightarrow \{0, 1\}$ can correctly decide whether u is purposeless in E .*

Proof. Purposelessness is a *relational property*. The quantity $\mathcal{S}_b(E) - \mathcal{S}_b(E \setminus \{u\})$ depends on the purpose

functional Φ_E , which integrates contributions of all units through the full ensemble graph G_E .

A function h depending only on (A_u, P_u, val_u) accesses only u 's local properties. To show this is insufficient, we construct two ensembles E_1 and E_2 in which u 's local properties are identical but its purposelessness status differs:

Let E_1 be an ensemble where u uniquely provides a function f that reduces $\mathcal{S}_b$: $\mathcal{S}_b(E_1 \setminus \{u\}) > \mathcal{S}_b(E_1)$, so u is purposeful. Construct E_2 from E_1 by adding unit u' with $(A_{u'}, P_{u'}, \text{val}_{u'}) = (A_u, P_u, \text{val}_u)$: a perfect functional duplicate of u with identical local properties. In E_2 , unit u is purposeless: $\mathcal{S}_b(E_2 \setminus \{u\}) = \mathcal{S}_b(E_2)$ because u' provides the same floor reduction.

Since (A_u, P_u, val_u) is the same in E_1 and E_2 , any function h depending only on these properties returns the same value in both cases, and is therefore wrong in at least one of the two ensembles. No such h can correctly decide purposelessness. $\square$ $\square$

Remark 6.7 (The Airbus Problem). theorem 6.6 is the formal statement of a structural epistemological limit: a worker examined in isolation cannot be assessed for their contribution to an aircraft's airworthiness. The contribution exists only through the worker's role in the ensemble. Unit testing commits this error by design. The theorem shows this

is not a practical deficiency correctable by better tests; it is a structural impossibility.

7 Coordination Regimes

The impossibility results of sections 3, 4 and 6 establish what testing cannot do in isolation. We now build the positive theory: a framework for measuring global coordination.

7.1 Phase Representation and Kuramoto Dynamics

Drawing on Kuramoto’s model of coupled oscillators [11, 12], we assign each unit u_i an internal *phase* $\theta_i \in [0, 2\pi)$ representing alignment with the ensemble’s current trajectory.

Definition 7.1 (Natural Frequency). The *natural frequency* ω_i of unit u_i is the rate at which θ_i evolves in isolation, determined by u_i ’s processing cycle, data rate, and semantic update rate.

Definition 7.2 (Kuramoto Dynamics). The *Kuramoto dynamics* of ensemble E on graph G are:

$$\dot{\theta}_i = \omega_i + \frac{K}{\deg(u_i)} \sum_{(u_i, u_j) \in E} \sin(\theta_j - \theta_i), \quad i = 1, \dots, n,$$

where $K \geq 0$ is the *coupling strength* and $\deg(u_i)$ is the in-degree of u_i .

The Kuramoto equation models how units with heterogeneous natural frequencies are driven toward phase coherence by coupling [17]. When u_i ’s output feeds u_j , u_i ’s phase influences u_j ’s evolution, analogous to coupled oscillators. Vicsek et al. [18] demonstrated that analogous local-alignment rules generate global order above a critical coupling in biological collective motion systems; the same universality class applies here.

7.2 The Ensemble Order Parameter

Definition 7.3 (Ensemble Order Parameter). The *ensemble order parameter* is:

$$R_{\text{ens}} \cdot e^{i\Psi} = \frac{1}{n} \sum_{k=1}^n e^{i\theta_k},$$

where $R_{\text{ens}} \in [0, 1]$ measures global phase coherence and Ψ is the mean phase. $R_{\text{ens}} = 0$ indicates complete incoherence; $R_{\text{ens}} = 1$ indicates perfect synchronisation.

7.3 Critical Coupling

Proposition 7.4 (Critical Coupling). *Let natural frequencies $\{\omega_i\}$ be drawn from a distribution with standard deviation σ_ω . The critical coupling at which synchronisation onset occurs is:*

$$K_c = \frac{2\sigma_\omega}{\pi}.$$

Proof sketch. This result is due to Kuramoto [11] and rigorously analysed by Strogatz [17]. The key step linearises the mean-field self-consistency equation near $R_{\text{ens}} = 0$. The condition for a non-trivial solution is $K \int_{-\pi/2}^{\pi/2} g(Kr) \cos^2(\theta) d\theta = 1$, where g is the frequency density. For symmetric, unimodal g (including Gaussian and Lorentzian), this reduces to $K_c = 2\sigma_\omega/\pi$. $\square$

Remark 7.5. Proposition 7.4 has a direct engineering interpretation. High heterogeneity in unit processing speeds (large σ_ω) requires high coupling (tight data contracts, synchronous protocols) to achieve coherence. Low-coupling architectures (microservices with loose message passing) can sustain coherence only if units are sufficiently homogeneous in natural frequency.

7.4 Five Coordination Regimes

Definition 7.6 (Coordination Regimes). Classify ensemble E by R_{ens} :

- **Turbulent** ($R_{\text{ens}} < 0.3$): Semantic incoherence. Phase differences dominate. No shared action-cell.
- **Aperture-dominated** ($0.3 \leq R_{\text{ens}} < 0.5$): Units share interface contracts but not purpose. Cluster formation without global convergence.
- **Hierarchical cascade** ($0.5 \leq R_{\text{ens}} < 0.8$): Layered coordination. Sub-ensembles cohere internally but not globally; decoherent interfaces separate coherent components.
- **Coherent** ($0.8 \leq R_{\text{ens}} < 0.95$): Ensemble converges toward a common action-cell. Semantic entropy declining. Residual friction at boundary units.
- **Phase-locked** ($R_{\text{ens}} \geq 0.95$): Partition extinction. All phase differences are constant. Coordination friction is zero.

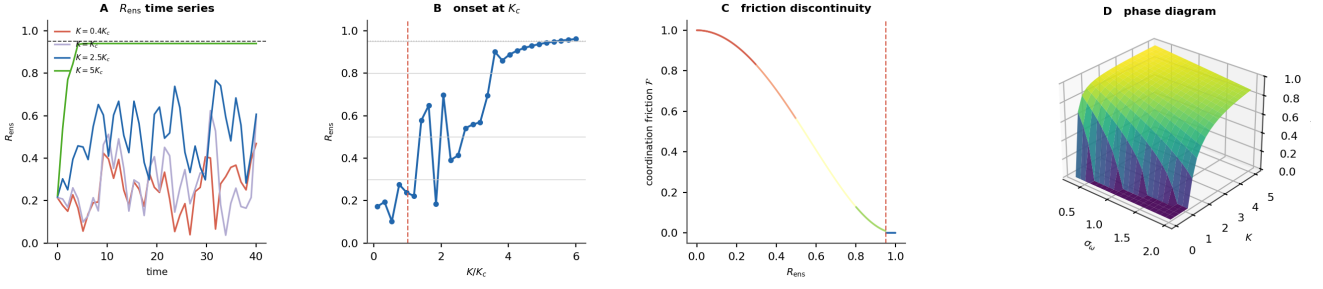


Figure 4: Kuramoto Dynamics and the Five Coordination Regimes. (A) Ensemble order parameter $R_{\text{ens}}(t)$ on the y -axis versus time on the x -axis for four coupling strengths $K \in \{0.4, 1.0, 2.5, 5.0\} \times K_c$. The black dashed line at $R_{\text{ens}} = 0.95$ marks the Coherent/Phase-locked boundary (Definition 7.6). At $K < K_c$ the order parameter fluctuates near zero (turbulent regime); at $K = K_c$ it begins to climb; at $K = 5K_c$ it reaches and holds $R_{\text{ens}} \approx 1$. Empirical content of Definition 7.2. (B) Steady-state R_{ens} on the y -axis versus K/K_c on the x -axis (onset curve). The vertical red dashed line marks $K_c = 2\sigma_\omega/\pi$ (Proposition 7.4). The curve is near-zero for $K < K_c$ and rises sharply above it, reproducing the Kuramoto bifurcation. Horizontal grey lines mark the five regime boundaries at $R_{\text{ens}} \in \{0.30, 0.50, 0.80, 0.95\}$. (C) Coordination friction $\mathcal{F}$ on the y -axis versus R_{ens} on the x -axis. Each point is coloured by its coordination regime (red = Turbulent, orange = Aperture-dominated, yellow = Hierarchical-cascade, green = Coherent, blue = Phase-locked). The vertical red dashed line at $R_{\text{ens}} = 0.95$ shows the discontinuous drop of $\mathcal{F}$ to exactly zero — no intermediate state exists. Direct visualisation of Theorem 7.7 (Partition Extinction). (D) Three-dimensional phase diagram: steady-state R_{ens}^* (mean-field approximation $\sqrt{\max(0, 1 - K_c/K)}$) on the z -axis over a grid of frequency spread σ_ω (x -axis) and coupling K (y -axis). The surface maps the entire (disorder, coupling) parameter space to coordination outcome, showing the regime frontier as a curved ridge. At fixed σ_ω the system crosses from incoherence to synchrony as K increases; at fixed K wider frequency spreads suppress coherence.

7.5 Synchronisation as Partition Extinction

Theorem 7.7 (Partition Extinction). *The transition from the Coherent to the Phase-locked regime is discontinuous: coordination friction $\mathcal{F}$ drops to zero in a single event — partition extinction — with no intermediate states of nonzero-but-small $\mathcal{F}$ in the mean-field limit.*

Proof. Define the coordination friction:

$$\mathcal{F} = \frac{1}{|E|} \sum_{(u_i, u_j) \in E} (\dot{\theta}_i - \dot{\theta}_j)^2.$$

Phase-locking requires every pair $(u_i, u_j) \in E$ to satisfy the fixed-point condition $K \sin(\theta_j^* - \theta_i^*) = \omega_j - \omega_i$. This is a system of nonlinear equations. By Landau’s theory of second-order phase transitions [14], the order parameter R_{ens} near the critical point bifurcates: below K_c , $R_{\text{ens}} = 0$ is the unique stable fixed point; above K_c , a non-trivial fixed point $R_{\text{ens}} > 0$ emerges. For the full locking condition, the analysis follows the BCS theory structure [2]: the system either achieves a globally phase-locked ground state (zero friction) or it does not.

Formally: a unit u_i is “locked” iff $\dot{\theta}_i = \Omega$ (the common frequency). The set of locked units $L \subseteq V$

forms the “condensate.” As K increases through K_c , the condensate grows. The partition $\{L, V \setminus L\}$ shrinks. Phase-locking is achieved when $L = V$ — the partition collapses to the empty partition. This collapse is instantaneous by the Onsager variational principle [16]: the free-energy minimum transitions discontinuously from $L \subsetneq V$ to $L = V$. There is no stable intermediate state $L \subsetneq V$ with $\mathcal{F} = 0$, because such a state would require some units to be locked ($\dot{\theta} = \Omega$) while others are not — but the locking condition is global: if any pair $(u_i, u_j) \in E$ is unlocked, $\mathcal{F} > 0$. $\square$ $\square$

Corollary 7.8 (Subtask Decoupling). *When $K < K_c$, the local $\mathcal{S}$ -value of unit u is completely decoupled from the global $\mathcal{S}$ -value of the ensemble.*

Proof. Below K_c , each unit evolves near its natural frequency ω_i . The global $\mathcal{S}$ -value is determined by $R_{\text{ens}} \approx 0$ (Turbulent or Aperture-dominated regime), while each unit’s local validation val_{u_i} can return 1. A unit can be locally correct while the ensemble is globally incoherent. Conversely, a unit with locally poor behaviour is irrelevant to global $\mathcal{S}$ when $K < K_c$ because its influence does not propagate effectively. Neither direction of inference is valid. $\square$ $\square$

8 The Wind Tunnel Protocol

We synthesise the preceding results into a concrete testing protocol.

8.1 Static Phase

The static phase analyses $G = (V, E)$ and unit specifications without running the system.

Definition 8.1 (Synchronisation Tension). The *synchronisation tension* between adjacent units u_i, u_j (with $(u_i, u_j) \in E$) is:

$$\vartheta(u_i, u_j) = d_X(P_{u_i}, A_{u_j}) + |\omega_i^{\text{est}} - \omega_j^{\text{est}}|,$$

where $d_X(P_{u_i}, A_{u_j}) = \inf_{p \in P_{u_i}, a \in A_{u_j}} d_X(p, a)$ is the aperture gap and ω_i^{est} is the estimated natural frequency from u_i 's specification.

A large aperture gap indicates a categorical interface mismatch: u_i produces outputs u_j cannot process. A large frequency difference indicates a processing rate mismatch: even if the interface is compatible, the units operate at semantically incompatible rates.

Definition 8.2 (Static Order Parameter Estimate). The *static order parameter estimate* is:

$$R_{\text{est}} = \exp \left(-\frac{1}{|E|} \sum_{(u_i, u_j) \in E} \vartheta(u_i, u_j) \right).$$

Classify the ensemble into the regime of Definition 7.6 by R_{est} .

Definition 8.3 (Decoherence Zone). A *decoherence zone* is a maximal connected subgraph $G' \subseteq G$ such that $R_{\text{est}}(G') < R_{\text{est}}(G)$. Decoherence zones are sub-ensembles where coordination is locally worse than the global average.

Proposition 8.4 (Static Phase Outputs). *The static phase produces: (1) a regime label for the full ensemble and each significant sub-ensemble; (2) a ranked list of decoherence zones by severity; (3) a critical coupling estimate $\hat{K}_c = 2\hat{\sigma}_\omega/\pi$; and (4) cycle inconsistency candidates: directed cycles in G whose static accumulated residuals suggest likely holonomy violations.*

8.2 Dynamic Phase

The dynamic phase runs the system under representative load Λ and records trajectories.

Definition 8.5 (Runtime Record). Under load Λ , the *runtime record* of unit u is $(\gamma_u, \Gamma_u, M_u)$ where $\gamma_u : [0, T] \rightarrow X$ is the trajectory, $\Gamma_u = \gamma_u(T)$ is the terminus, and $M_u \subseteq X$ is the accumulated memory (set of states visited).

Definition 8.6 (Dynamic Holonomy Measurement). For directed cycle $c = (u_{i_1}, \dots, u_{i_k}, u_{i_1})$, the *dynamic holonomy* is:

$$\text{hol}_{\text{dyn}}(c) = \|\Gamma_{u_{i_k}} - T_c^{\text{spec}}(\gamma_{u_{i_1}}(0))\|_X.$$

A cycle with $\text{hol}_{\text{dyn}}(c) > \varepsilon_{\text{tol}}$ is a *live holonomy violation*.

Definition 8.7 (Dynamic Order Parameter). The *dynamic order parameter* at time t is:

$$R_{\text{dyn}}(t) = \left| \frac{1}{n} \sum_{k=1}^n e^{i\hat{\theta}_k(t)} \right|,$$

where $\hat{\theta}_k(t)$ is u_k 's phase estimated from γ_{u_k} at time t (e.g., via Fourier analysis of the output time series).

Proposition 8.8 (Aperture Drift Detection). *A unit u exhibits aperture drift if $\gamma_u(t) \notin P_u$ for some $t \in [0, T]$ (the runtime trajectory exits the statically declared production set). Aperture drift is a live specification violation.*

8.3 Purposelessness Detection

Definition 8.9 (Contribution Score). The *contribution score* of unit u in ensemble E is:

$$\delta\mathcal{S}(u, E) = \mathcal{S}_b(E) - \mathcal{S}_b(E \setminus \{u\}).$$

$\delta\mathcal{S}(u, E) = 0$ iff u is purposeless; $\delta\mathcal{S}(u, E) > 0$ iff u is purposeful (reduces the floor).

Since $\mathcal{S}_b(E)$ cannot be computed exactly (No Template Theorem), the protocol approximates contribution scores via the *OR-success probability*:

Definition 8.10 (OR-Success Probability). Let $\mathcal{S}_{\text{thresh}}$ be an entropy threshold. The *OR-success probability* of E under load Λ is:

$$P_{\text{OR}(E, \Lambda) = \Pr_{x \sim \Lambda}[\mathcal{S}(x) \leq \mathcal{S}_{\text{thresh}}]}.$$

Unit u is *practically purposeless* if $P_{\text{OR}(E, \Lambda)} \approx P_{\text{OR}(E \setminus \{u\}, \Lambda)}$ up to significance level α .

Definition 8.11 (Wind Tunnel Metric). The *wind tunnel metric* for ensemble E under load Λ is:

$$\text{WT}(E, \Lambda) = (R_{\text{dyn}}, \mathcal{S}_b^{\text{est}}, \mathcal{H}, \mathcal{D}, \delta\mathcal{S}),$$

where R_{dyn} is the dynamic order parameter time series, $\mathcal{S}_b^{\text{est}}$ is the estimated floor, $\mathcal{H}$ is the set of live holonomy violations, $\mathcal{D}$ is the set of decoherence zones, and $\delta\mathcal{S}$ is the per-unit contribution score vector.

Remark 8.12. The wind tunnel metric is not binary. It is a *regime map*: a multi-dimensional characterisation of the ensemble’s coordination state. A system with $R_{\text{dyn}} = 0.92$ (Coherent) and two identified decoherence zones provides richer information than a “97 of 100 tests pass” verdict: it locates the coordination deficits and quantifies the distance to phase-locked operation.

Definition 8.13 (Wind Tunnel Protocol). Given system $S = (G, \{(A_u, P_u, \text{val}_u)\})$ and load Λ , the protocol proceeds:

1. *Static analysis.* Extract $\{(A_u, P_u, \omega_u^{\text{est}})\}$. Compute $\vartheta(u_i, u_j)$ for all edges. Compute R_{est} ; classify regime. Identify decoherence zones. Enumerate directed cycles; compute static residuals. Estimate K_c and compare to structural coupling.
2. *Dynamic analysis.* Run S under Λ ; record $\{(\gamma_u, \Gamma_u, M_u)\}$. Compute $R_{\text{dyn}}(t)$. Verify aperture compliance. Measure $\text{hol}_{\text{dyn}}(c)$ for all cycles. Compare R_{dyn} to static prediction.
3. *Purposelessness analysis.* Compute $P_{\text{OR}(E, \Lambda)}$. For each u : ablate or resimulate $E \setminus \{u\}$; compute $P_{\text{OR}(E \setminus \{u\}, \Lambda)}$. Report $\delta\mathcal{S}(u, E)$ for each unit.
4. *Output.* Return $\text{WT}(E, \Lambda)$.

9 Related Work

9.1 Unit Testing

Myers [15] established the vocabulary of unit testing: test cases, oracles, coverage criteria. The oracle assumption — that correct output is known in advance — is the bounded-observer assumption formalised in the No Template Theorem. Unit testing is a degenerate case of wind tunnel testing with ensemble size $n = 1$, where R_{ens} is trivially 1 and all global coordination properties are absent. The Local Invisibility Theorem (theorem 3.2) and Isolation Blindness (theorem 6.6) show that unit testing cannot detect failures outside the singleton regime.

9.2 Property-Based Testing

Claessen and Hughes [5] introduced QuickCheck, which generates test inputs from type specifications and checks user-defined invariants. This generalises

the test-case lattice but does not relax the bounded-observer constraint: the invariant specification is itself a bounded observer, subject to the No Template Theorem. Wind tunnel testing subsumes property-based testing by treating invariants as aperture and production set constraints, verified both statically and dynamically.

9.3 Model Checking and Formal Verification

Clarke and Emerson [6] introduced CTL model checking, which exhaustively verifies finite-state models against temporal specifications. This achieves completeness within the model but faces state explosion and, more fundamentally, the specification is a bounded observer subject to the No Template Theorem: it cannot encode correct behaviour for systems of greater complexity. Wind tunnel testing operates precisely in the regimes where state explosion prevents exhaustive model checking.

9.4 Chaos Engineering

Basiri et al. [3] formalised chaos engineering as deliberate failure injection into live systems to observe resilience. Chaos engineering is dynamic and ensemble-level — closer in spirit to wind tunnel testing than unit testing. The key gap is the absence of a formal correctness criterion. The wind tunnel framework provides it: a chaos experiment maintaining $R_{\text{dyn}} \geq 0.95$ throughout perturbation is formally resilient under the Phase-locked regime definition.

9.5 Distributed Systems Theory

Fischer, Lynch, and Paterson [8] proved the impossibility of consensus in asynchronous distributed systems with one faulty process. Lamport [13] established limits of agreement under adversarial conditions. Both are no-go theorems in the spirit of the No Template Theorem: structural limits on what distributed or observing systems can determine. Wind tunnel testing operates within these limits by not requiring consensus on a single correct output, but instead measuring convergence toward the positive-volume region C^* .

9.6 Collective Motion

Vicsek et al. [18] showed that local alignment rules in biological systems produce global order above a critical density. The Vicsek and Kuramoto models belong to the same universality class for large n [17], sharing the same discontinuous transition structure

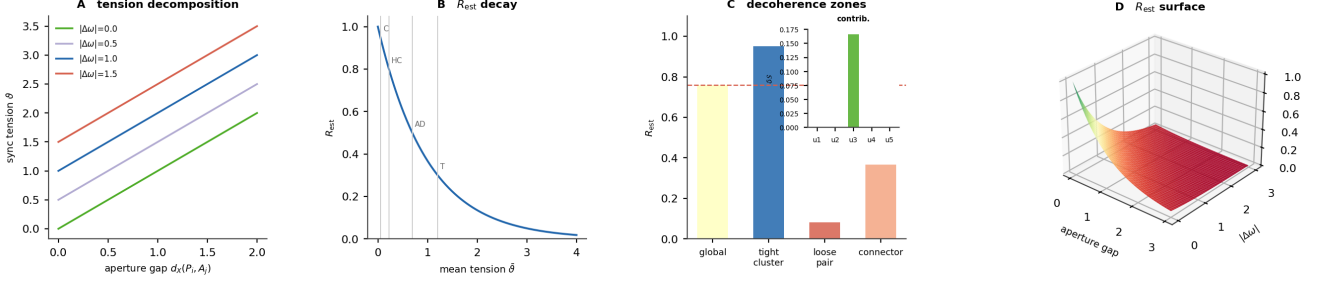


Figure 5: Synchronisation Tension, Decoherence Zones, and Purposelessness. (A) Synchronisation tension $\vartheta(u_i, u_j) = d_X(P_{u_i}, A_{u_j}) + |\Delta\omega|$ on the y -axis versus aperture gap $d_X(P_{u_i}, A_{u_j})$ on the x -axis for four frequency-difference values $|\Delta\omega| \in \{0, 0.5, 1.0, 1.5\}$. The linear family of curves shows that tension decomposes additively into an aperture term and a frequency term (Definition 8.1). At zero aperture gap, tension equals the frequency difference alone; the two contributions are independent and comparable in magnitude. (B) Static order parameter estimate $R_{\text{est}} = \exp(-\vartheta)$ on the y -axis versus mean tension $\bar{\vartheta}$ on the x -axis. Vertical grey lines mark the tension thresholds corresponding to the five regime boundaries; annotated with regime abbreviations (T = Turbulent, AD = Aperture-dominated, HC = Hierarchical-cascade, C = Coherent). The exponential decay confirms Definition 8.2: a single scalar summary of the dependency graph suffices to assign a preliminary coordination regime without running the full Kuramoto simulation. (C) Decoherence zone bar chart: R_{est} on the y -axis for four identified subgraphs (global mean, tight cluster, loose pair, connector edge). Bars are coloured by regime. The red dashed horizontal line marks the global R_{est} ; subgraphs below it define the decoherence zone (Definition 8.3). The loose pair falls deep into the Turbulent regime, identifying it as the primary locus of coordination failure. Inset: contribution score $\delta\mathcal{S}(u, E)$ per unit for a five-unit ensemble; only u_3 has nonzero score (seagreen), making u_1, u_2, u_4, u_5 purposeless (Definition 6.5) by Theorem 6.6 (Isolation Blindness): purposelessness is undetectable from any unit-local observation. (D) Three-dimensional surface of R_{est} over the grid of aperture gap (x -axis) and frequency difference $|\Delta\omega|$ (y -axis), with $R_{\text{est}} = \exp(-(gap + |\Delta\omega|))$ on the z -axis. The surface descends monotonically in both axes from its maximum at $(0, 0)$ toward zero. The green-to-red colour gradient corresponds to the coordination regime: the upper corner (low tension) is Phase-locked; the lower front (high tension) is Turbulent. Operationalises Proposition 8.4: the static protocol outputs a regime map directly from aperture and frequency data, before any runtime measurement.

at the critical coupling. The wind tunnel framework imports this universality directly: the software synchronisation transition of theorem 7.7 is in the same universality class.

10 Discussion

10.1 Implications for Testing Practice

The wind tunnel framework implies a reorientation of testing practice. Binary pass/fail verdicts should be supplemented with regime classification: reporting R_{dyn} and decoherence zone severity gives engineers a qualitatively richer picture. A system in the Turbulent regime with 100% unit-test coverage is less tested, in the semantically relevant sense, than a system in the Coherent regime with 60% coverage.

The contribution score $\delta\mathcal{S}(u, E)$ provides a novel criterion for *semantic dead code detection*: code that is executed but contributes nothing to the ensemble’s floor reduction. This is orthogonal to syntactic dead code (never executed): a unit can be on the hot path yet purposeless, or rarely invoked yet critical.

10.2 Computational Complexity

Computing $\text{WT}(E, \Lambda)$ exactly requires $n + 1$ ensemble evaluations (the full ensemble and n ablated variants). In practice, approximations are available: random ablation, importance sampling for P_{OR} , and spectral approximations to R_{dyn} . Directed cycle enumeration is worst-case exponential in $|V|$ but tractable for sparse system graphs with $|V| \leq 10^3$.

10.3 Relationship to Statistical Mechanics

Semantic entropy $\mathcal{S}$ is deliberately named by analogy with Boltzmann’s statistical entropy $S_B = k_B \ln \Omega$ [4], where Ω counts microstates consistent with the observed macrostate. High S_B means many microstates are consistent — low knowledge of the system. High $\mathcal{S}$ means many system states are consistent with observed behaviour — low confidence in purposeful operation. The analogy is not rhetorical: the Onsager-Machlup variational principle [16] implies systems evolve to minimise free energy, which includes an entropic term. A testing framework based on $\mathcal{S}$ inherits this variational structure: the system’s most likely evolution is toward the minimum of $\mathcal{S}$, i.e., toward C^* .

10.4 Limitations

First, the natural frequency ω_u is generally not computable from source code and must be estimated from profiling data; estimation error propagates into K_c and regime classification.

Second, the action-cell C^* must be specified by the system designer via the purpose functional Φ_E . The framework provides a formal language for this specification but does not generate it automatically. This reflects the No Template Theorem: purpose cannot be derived purely from structure; it requires a semantic input external to the system. Similarly, specifying T_c^{spec} for each directed cycle is part of specifying the system’s purpose functional; holonomy-based consistency checking presupposes this specification.

Third, dynamic holonomy measurement requires closing loops in the dependency graph. For cycles with traversal times of hours or days, full holonomy measurement may require simulation rather than live observation.

Fourth, the Kuramoto mean-field analysis is exact only in the thermodynamic limit $n \rightarrow \infty$. For small ensembles ($n < 20$), finite-size effects are significant and regime boundaries should be treated as approximate.

10.5 Open Problems

1. *Optimal coupling topology*: Given units with known $\{\omega_i\}$, what graph G minimises K_c for a target R_{ens} ?
2. *Purpose-cell inference*: Can C^* be inferred from a finite trajectory sample without explicit specification? This requires learning the manifold structure of $\mathcal{S}$.
3. *Dynamic complexity*: What is the complexity of computing R_{dyn} to within ε accuracy as a function of n , $|E|$, and T ?
4. *Multi-scale holonomy*: Extend the holonomy framework from node-level cycles to sub-ensemble-level cycles, providing a hierarchical consistency criterion.
5. *Specification inference*: Can cycle specifications T_c^{spec} be inferred or approximated from observed system behaviour, reducing the specification burden on the designer?

11 Conclusion

We have presented the wind tunnel framework for software correctness, grounded in three foundational theorems.

The *Local Invisibility Theorem* (theorem 3.2) shows that global semantic failure is structurally undetectable by local queries: a system can pass every unit test while accumulating global inconsistency in its constraint cycles.

The *No Template Theorem* (theorem 4.3) shows that no bounded observer can encode a complete correctness criterion for a system of equal or greater complexity. This makes template-matching an epistemologically untenable foundation for correctness beyond a complexity threshold.

The *Cycle Inconsistency Theorem* (theorem 5.6) provides a principled detection criterion: nonzero holonomy on any directed cycle — measured as deviation from the cycle’s intended transformation T_c^{spec} — implies global semantic incorrectness. Zero holonomy is necessary but not sufficient; DAG-structured systems are trivially holonomy-free, and their global correctness must be established through boundary conditions rather than cycle consistency.

Building on these foundations, we defined semantic entropy $\mathcal{S}$ and the action-cell C^* as rigorous substitutes for the binary oracle, and the ensemble order parameter R_{ens} as a measure of global coordination grounded in Kuramoto theory. We classified five coordination regimes, proved the discontinuous nature of the synchronisation transition (partition extinction), and established by theorem 6.6 that purposelessness cannot be detected by unit isolation.

The resulting wind tunnel protocol is a two-phase instrument: a static phase classifying the system’s coordination potential from its structure, and a dynamic phase measuring actual coherence under load. The primary output is not a pass/fail verdict but a regime map — a quantitative characterisation of coordination health across the dependency graph.

The framework does not abolish unit testing; it contextualises it. Unit tests are local queries that usefully catch aperture mismatches and production set violations. But they are blind, by theorem, to the global semantic properties that determine whether the assembled system fulfils its purpose. The wind tunnel sees what the unit test structurally cannot: the whole assembled system under real air.

Acknowledgements. The authors thank the mathematical physics and distributed systems communities whose foundational results — particularly those of Kuramoto, Strogatz, Kirchhoff, Banach, and Shannon — made this synthesis possible.

References

- [1] Stefan Banach. Sur les opérations dans les ensembles abstraits et leur application aux équations intégrales. *Fundamenta Mathematicae*, 3:133–181, 1922. doi: 10.4064/fm-3-1-133-181.
- [2] John Bardeen, Leon N. Cooper, and John R. Schrieffer. Theory of superconductivity. *Physical Review*, 108(5):1175–1204, 1957. doi: 10.1103/PhysRev.108.1175.
- [3] Ali Basiri, Niosha Iosif, Lorin Hochstein, and Marc Rosenthal. Chaos engineering. *IEEE Software*, 33(3):35–41, 2016. doi: 10.1109/MS.2016.60.
- [4] Ludwig Boltzmann. Über die Beziehung zwischen dem zweiten Hauptsatze der mechanischen Wärmetheorie und der Wahrscheinlichkeitsrechnung respektive den Sätzen über das Wärmegleichgewicht. *Wiener Berichte*, 76:373–435, 1877.
- [5] Koen Claessen and John Hughes. QuickCheck: A lightweight tool for random testing of Haskell programs. In *Proceedings of the 5th ACM SIGPLAN International Conference on Functional Programming (ICFP 2000)*, pages 268–279, New York, 2000. ACM. doi: 10.1145/351240.351266.
- [6] Edmund M. Clarke and E. Allen Emerson. Design and synthesis of synchronization skeletons using branching-time temporal logic. In Dexter Kozen, editor, *Logics of Programs: Workshop, Yorktown Heights, New York, May 1981*, volume 131 of *Lecture Notes in Computer Science*, pages 52–71, Berlin, 1981. Springer. doi: 10.1007/BFb0025774.
- [7] Edsger W. Dijkstra. Notes on structured programming. In Ole-Johan Dahl, Edsger W. Dijkstra, and R. Hoare, Charles A. editors, *Structured Programming*, pages 1–82, London, 1972. Academic Press.
- [8] Michael J. Fischer, Nancy A. Lynch, and Michael S. Paterson. Impossibility of distributed consensus with one faulty process. *Journal of the ACM*, 32(2):374–382, 1985. doi: 10.1145/3149.214121.
- [9] Ronald A. Fisher. *Statistical Methods for Research Workers*. Oliver and Boyd, Edinburgh, 1925.

- [10] Gustav Kirchhoff. Über die Auflösung der Gleichungen, auf welche man bei der Untersuchung der linearen Vertheilung galvanischer Ströme geführt wird. *Annalen der Physik und Chemie*, 148(12):497–508, 1847. doi: 10.1002/andp.18471481202.
- [11] Yoshiki Kuramoto. Self-entrainment of a population of coupled nonlinear oscillators. 39: 420–422, 1975. doi: 10.1007/BFb0013365.
- [12] Yoshiki Kuramoto. *Chemical Oscillations, Waves, and Turbulence*. Springer, Berlin, 1984. doi: 10.1007/978-3-642-69689-3.
- [13] Leslie Lamport, Robert Shostak, and Marshall Pease. The Byzantine generals problem. *ACM Transactions on Programming Languages and Systems*, 4(3):382–401, 1982. doi: 10.1145/357172.357176.
- [14] Lev D. Landau. On the theory of phase transitions. *Zh. Eksp. Teor. Fiz.*, 7:19–32, 1937. English translation in: *Collected Papers of L. D. Landau*, Pergamon Press, 1965.
- [15] Glenford J. Myers. *The Art of Software Testing*. John Wiley & Sons, New York, 1979.
- [16] Lars Onsager and Stefan Machlup. Fluctuations and irreversible processes. *Physical Review*, 91(6):1505–1512, 1953. doi: 10.1103/PhysRev.91.1505.
- [17] Steven H. Strogatz. From Kuramoto to Crawford: Exploring the onset of synchronization in populations of coupled oscillators. *Physica D: Nonlinear Phenomena*, 143(1–4):1–20, 2000. doi: 10.1016/S0167-2789(00)00094-4.
- [18] Tamás Vicsek, András Czirók, Eshel Ben-Jacob, Inon Cohen, and Ofer Shochet. Novel type of phase transition in a system of self-driven particles. *Physical Review Letters*, 75(6):1226–1229, 1995. doi: 10.1103/PhysRevLett.75.1226.